

CONSTRUCTION DOCUMENTATION

Design Title : Machine Stand
Revision Level :
Approved by :
Designer : Frame Expert
Company : FX
Address : Ottawa, ON
E-Mail : support@framexpert.com
Phone :
Date : 2025-06-19



Bill of materials (Number of kits: 1)

Pos	Article-No	Description	Units	Qty
1	1.11.040040.43LP-A00A00/208	Profile 40x40, 4E, light, plain	MM	1
2	1.11.040040.43LP-A00A00/208	Profile 40x40, 4E, light, plain **PER DRAWING 1**	MM	1
2.A	Q11A4I07161-5.5	Counterbore, left Ø 5.5 mm, distance 14.0 mm, Side: 4	EA	1
2.B	Q11A4I07161-5.5	Counterbore, left Ø 5.5 mm, distance 194.0 mm, Side: 4	EA	1
3	1.11.040040.43LP-A00A00/328	Profile 40x40, 4E, light, plain	MM	1
4	1.11.040040.43LP-A00A00/328	Profile 40x40, 4E, light, plain **PER DRAWING 2**	MM	1
4.A	Q11A4I07161-5.5	Counterbore, left Ø 5.5 mm, distance 74.0 mm, Side: 4	EA	1
4.B	Q11A4I07161-5.5	Counterbore, left Ø 5.5 mm, distance 254.0 mm, Side: 4	EA	1
5	1.11.040040.43LP-AA1AA2/460	Profile 40x40, 4E, light, plain	MM	1
6	1.11.040040.43LP-AA2AA2/460	Profile 40x40, 4E, light, plain	MM	1
7	1.11.040040.43LP-AA4AA2/460	Profile 40x40, 4E, light, plain	MM	1
8	1.11.040040.43LP-AA4AA2/610	Profile 40x40, 4E, light, plain	MM	1
9	1.11.040040.43LP-AL0A00/700	Profile 40x40, 4E, light, plain Depth of thread, left: 16.0 mm (M14)	MM	1
10	1.11.040040.43LP-AA2AA2/610	Profile 40x40, 4E, light, plain	MM	2
11	1.11.040040.43LP-AA4AA4/460	Profile 40x40, 4E, light, plain	MM	3
12	1.11.040040.43LP-AA4AA4/610	Profile 40x40, 4E, light, plain	MM	3
13	1.11.040040.43LP-A00AL0/700	Profile 40x40, 4E, light, plain Depth of thread, right: 16.0 mm (M14)	MM	3
14	1.11.040040.43LP-AA4AA4/520	Profile 40x40, 4E, light, plain	MM	4
15	1.41.E314.1	Combination profile PP, E, 1 - 4.5, grey (bar 2.5 m)	EA	3
22	1.42.20404.2	Cover cap 40x40, dia: 12, black	EA	8
23	1.45.31075	Swivel castor stop fix with bolt hole fitting, dia: 75	EA	4
24	1.46.115	Angle 25x40, dia: 8.7	EA	4
25	1.61.210	Handle light	EA	2
26	1.65.2538078	Lock GD-ZN	EA	2
27	0.63.D00912.08016	Cylindric head screw, DIN 912 - M8x16	EA	4
28	0.63.D06912.05035	Cylindric head screw, DIN 6912 - M5x35	EA	4
29	0.63.D06912.06014	Cylindric head screw, DIN 6912 - M6x14	EA	4
30	0.63.D06912.06016	Cylindric head screw, DIN 6912 - M6x16	EA	8
31	0.63.D06912.10030	Cylindric head screw, DIN 6912 - M10x30	EA	4
32	0.63.D07991.04012	Flat head screw, DIN 7991 - M4x12	EA	49
33	0.63.D07991.06014	Flat head screw, DIN 7991 - M6x14	EA	12
34	1.21.4E0	Connector, universal	EA	32
35	0.61.D06923.08	Hexagonal nut with flange DIN 6923 - M8	EA	4
36	1.31.EM4	Threaded plate E, M4	EA	49
37	1.32.4EM6	T-Nut for subsequent insertion, w. spring E, M6	EA	24
38	1.35.1141015	Threaded insert M14/M10, L15	EA	4

Bill of materials (Number of kits: 1)

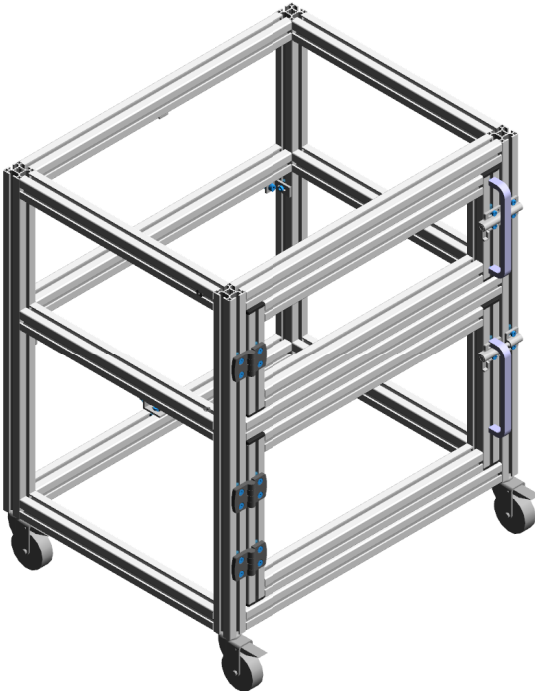
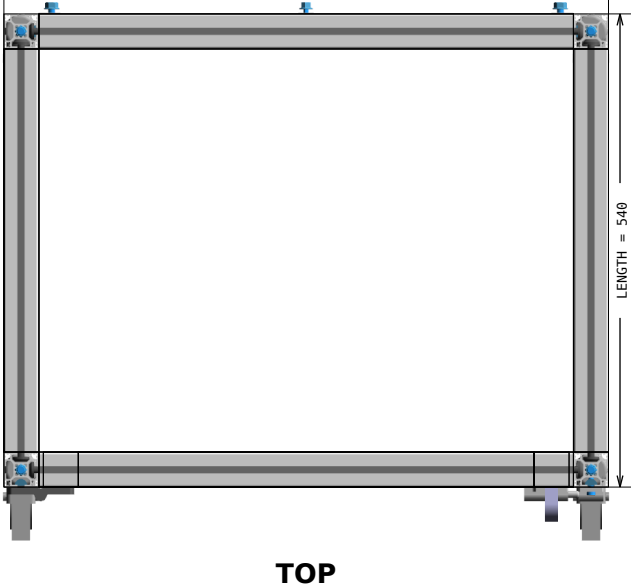
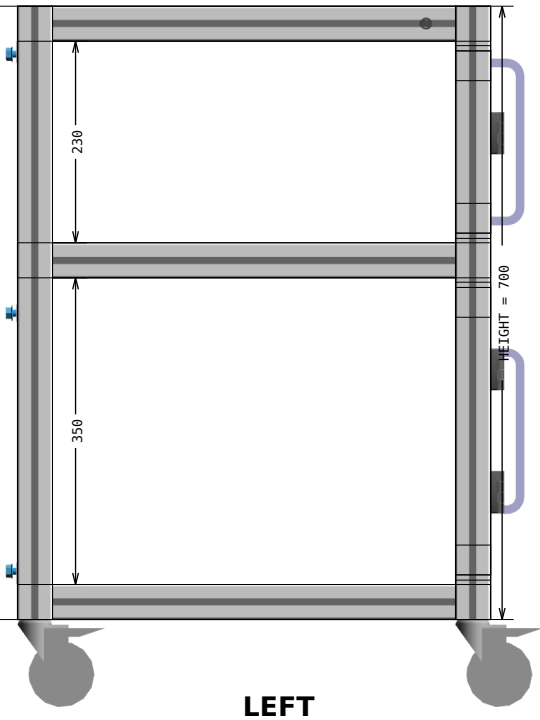
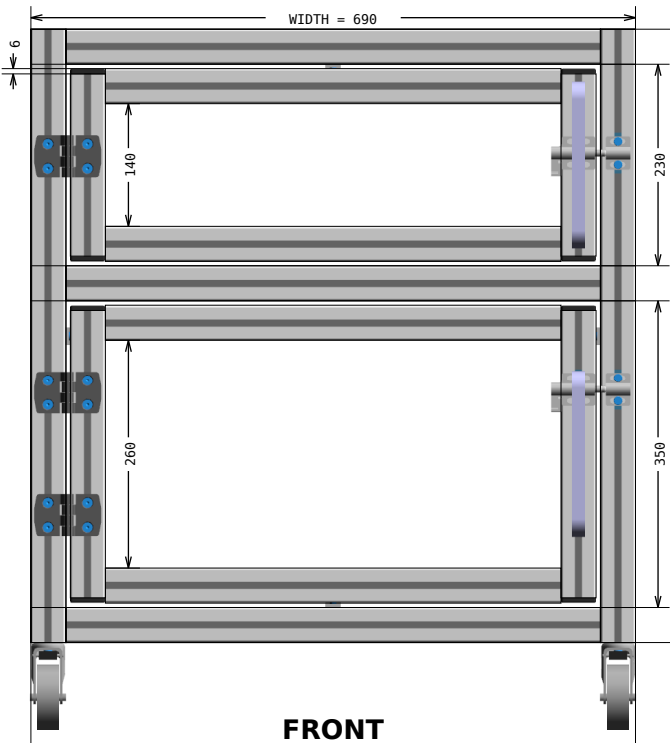
Pos	Article-No	Description	Units	Qty
39	1.62.448.22/22	Hinge 48 fixed, A 22.5	EA	3
		Frame weight: 23.20 kg (51.16 lbs)		
16	2.87.1571-99	Polycarbonat 4mm, SIZE: 160.00MM x 540.00MM **PER DRAWING 3**	SM	1
17	2.87.1571-99	Polycarbonat 4mm, SIZE: 280.00MM x 540.00MM **PER DRAWING 4**	SM	1
18	2.87.1571-99	Polycarbonat 4mm, SIZE: 480.00MM x 630.00MM **PER DRAWING 5**	SM	1
19	2.87.1571-99	Polycarbonat 4mm, SIZE: 540.00MM x 690.00MM **PER DRAWING 6**	SM	1
20	2.87.1571-99	Polycarbonat 4mm, SIZE: 690.00MM x 700.00MM **PER DRAWING 7**	SM	1
21	2.87.1571-99	Polycarbonat 4mm, SIZE: 540.00MM x 700.00MM **PER DRAWING 8**	SM	2
		Panels weight: 9.24 kg (20.37 lbs)		
		Total Weight: 32.45 kg (71.53 lbs)		

Isometric view

Some accessories may not be shown

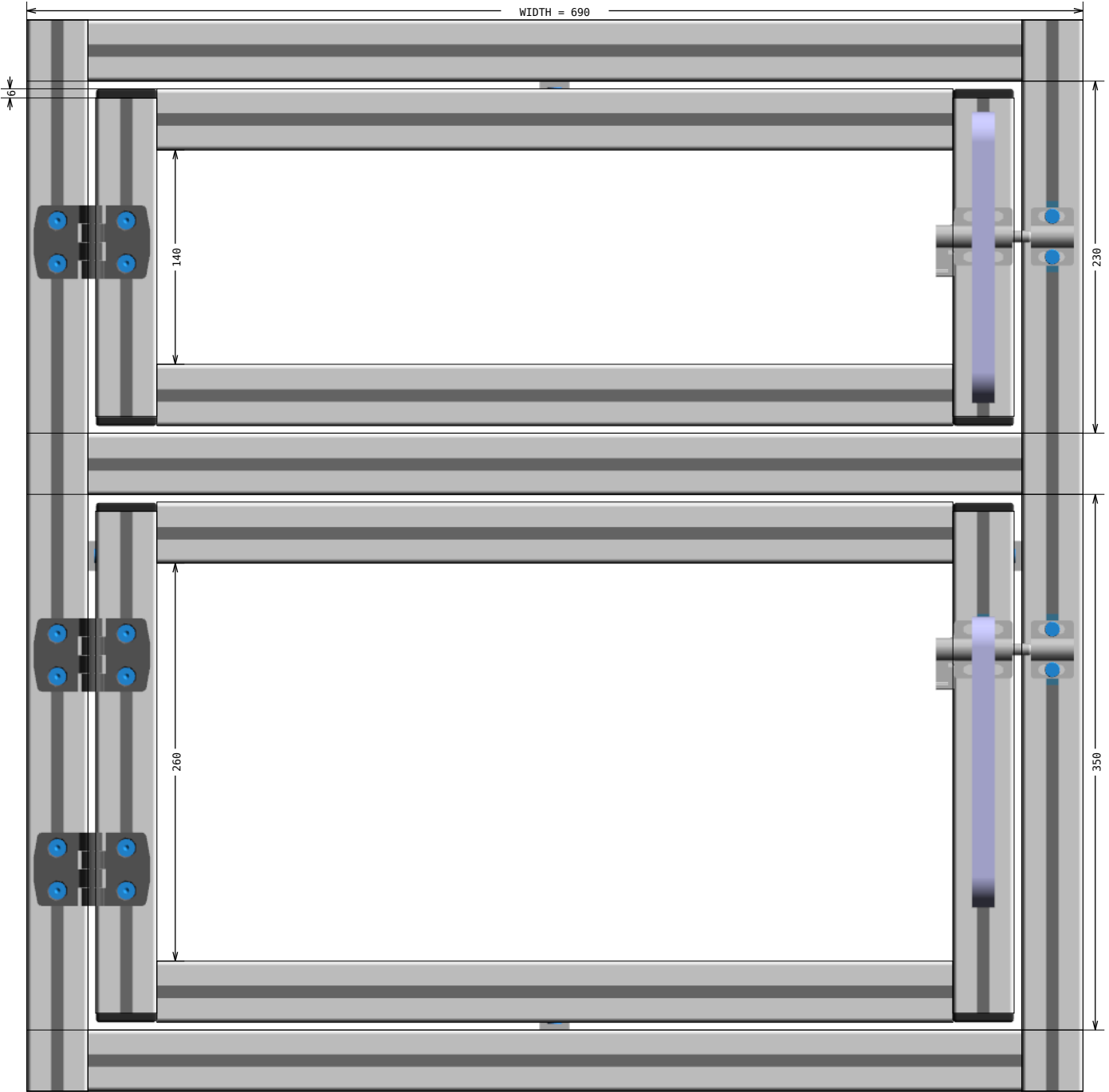


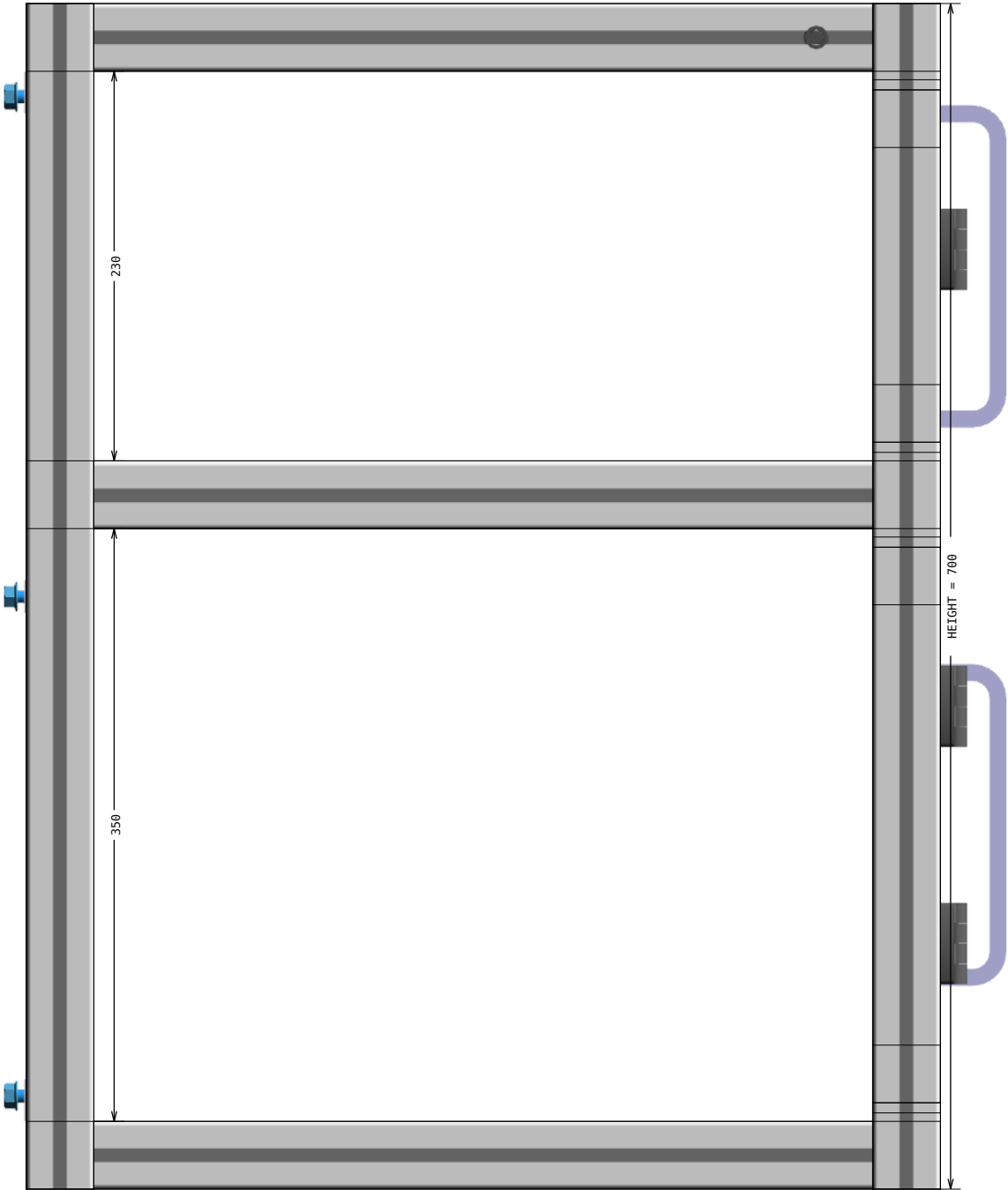
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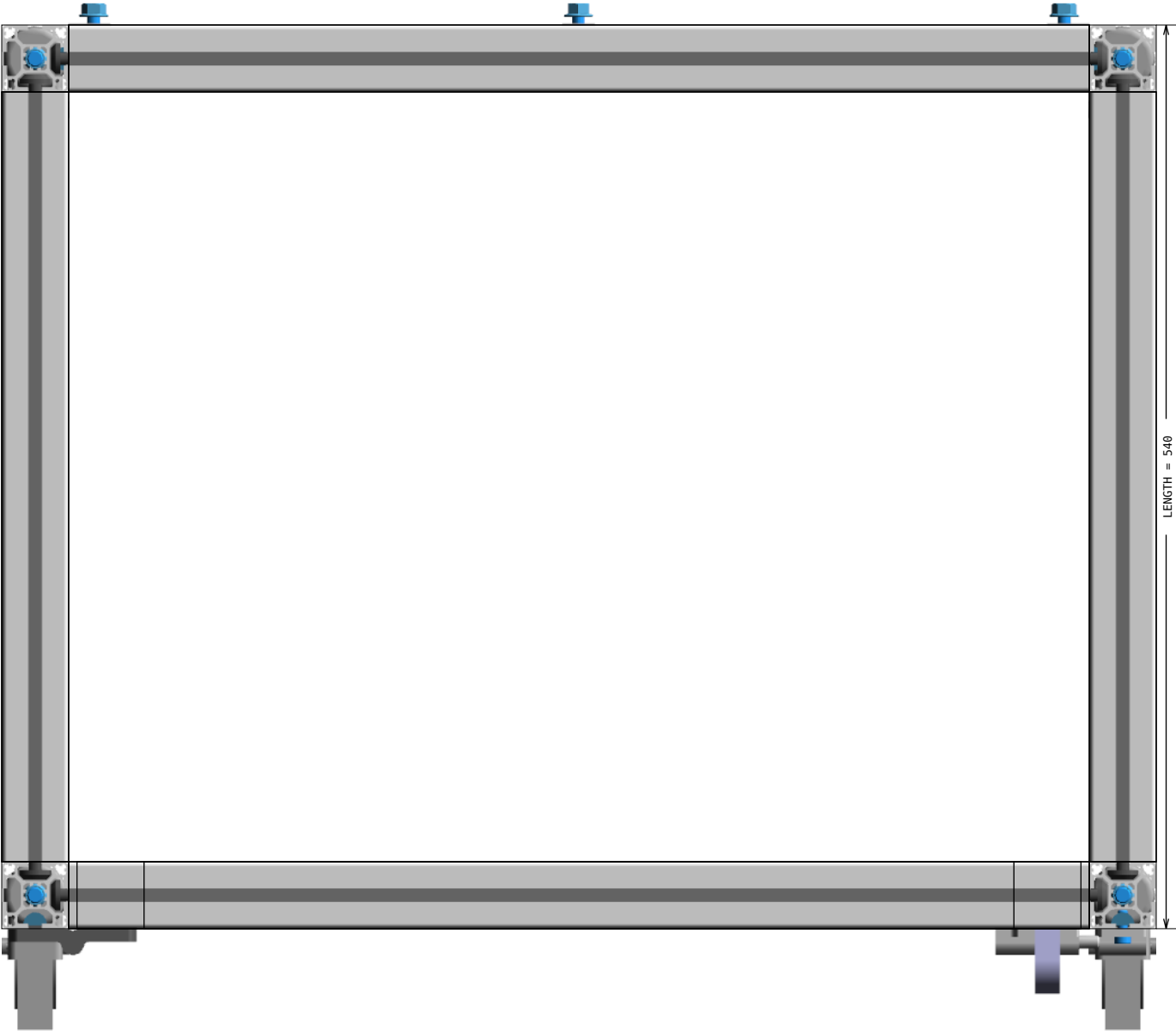


ISOMETRIC

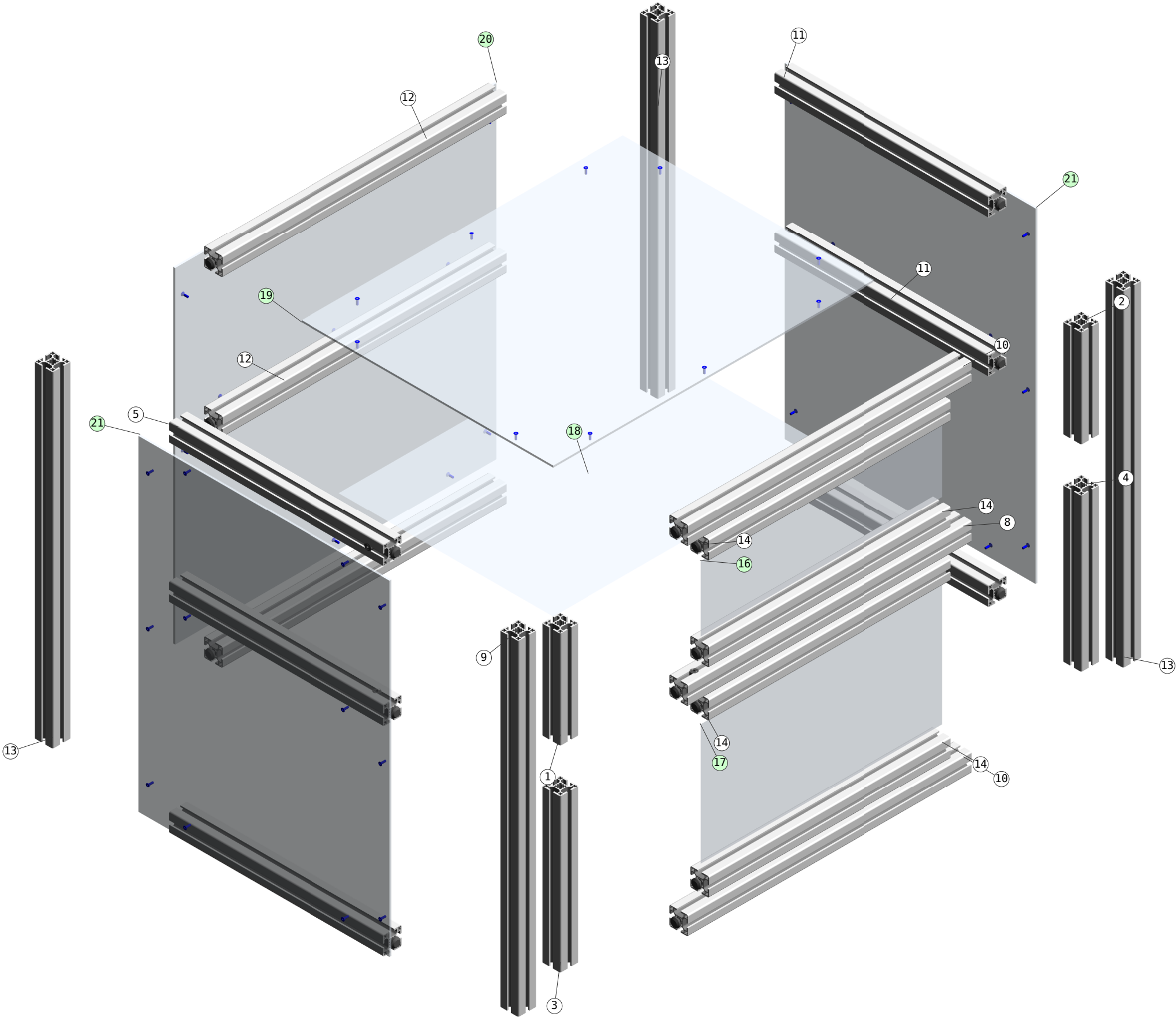
FRONT VIEW

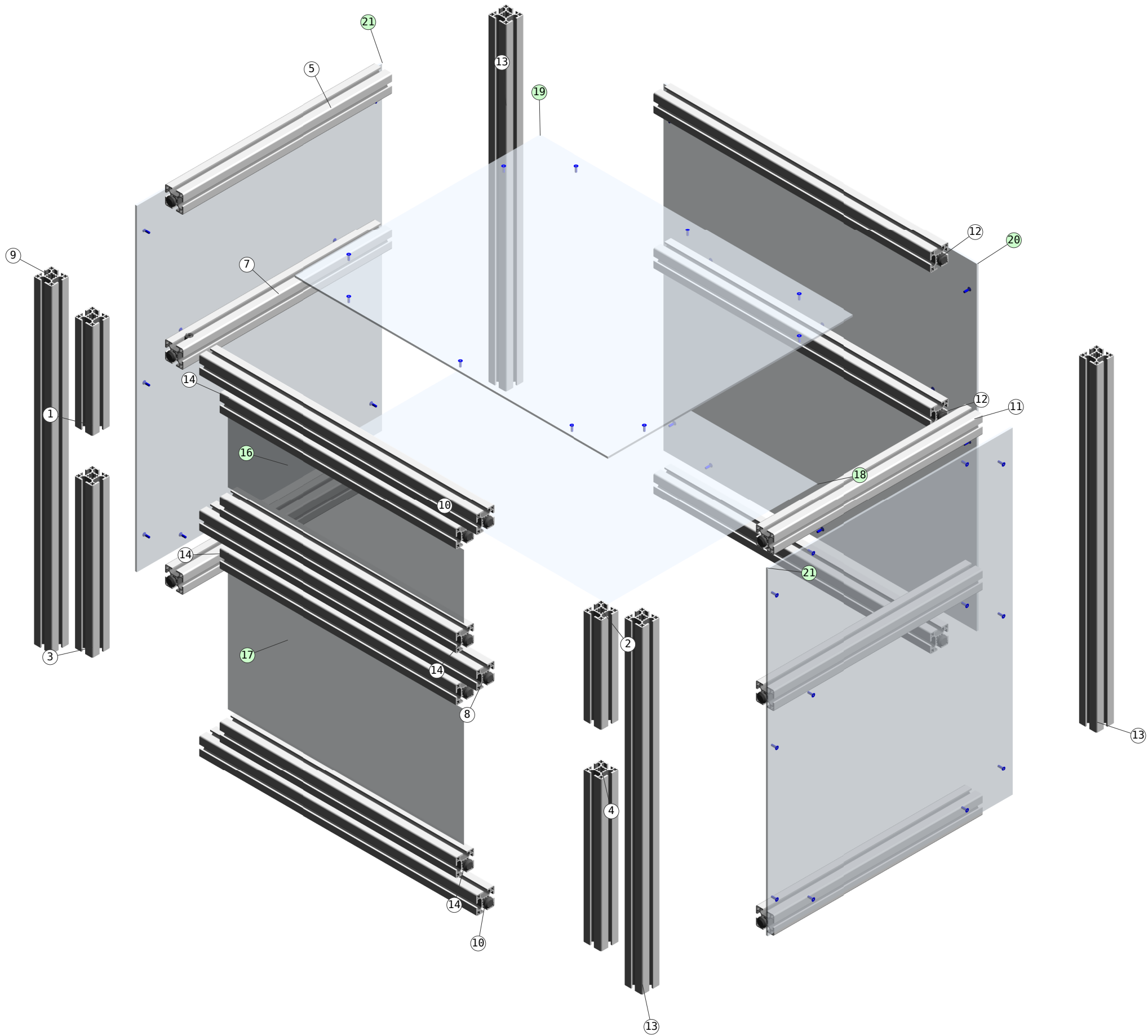






Exploded view





Components of group: Door
Panels are not shown for image clarity, their locations are marked by green balloons



Components of group: Door(1)

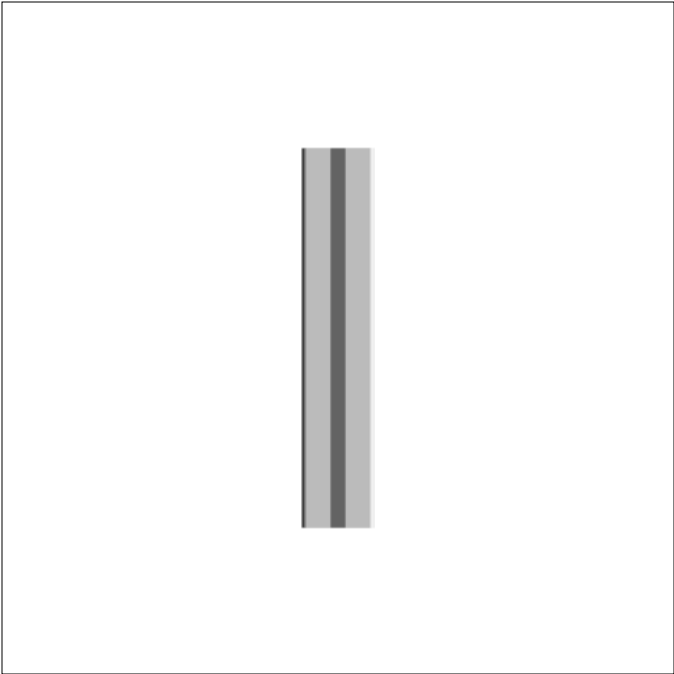
Panels are not shown for image clarity, their locations are marked by green balloons



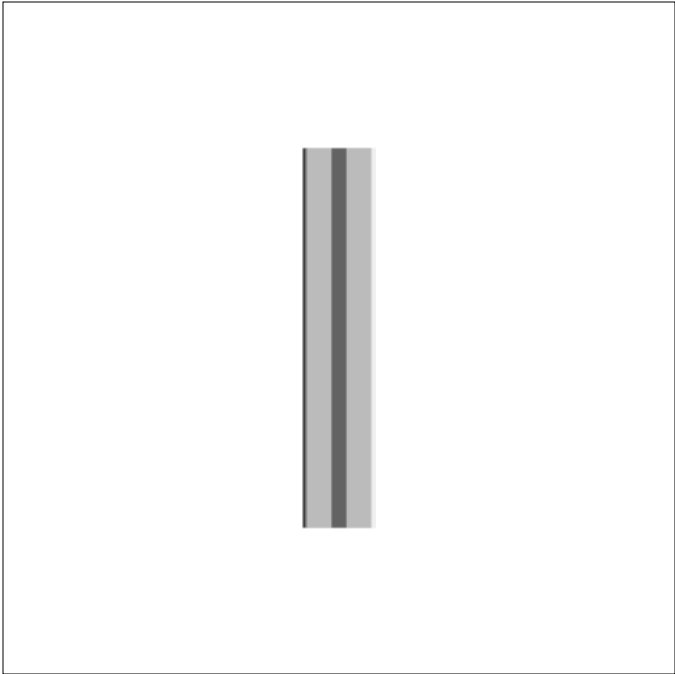
Assembly List

Pos	Article-No	Description	Related parts	Qty
5	1.11.040040.43LP-AA1AA2/460	Profile 40x40, 4E, light, plain	1.21.4E0, Connector, universal	2
6	1.11.040040.43LP-AA2AA2/460	Profile 40x40, 4E, light, plain	1.21.4E0, Connector, universal	2
7	1.11.040040.43LP-AA4AA2/460	Profile 40x40, 4E, light, plain	1.21.4E0, Connector, universal	2
8	1.11.040040.43LP-AA4AA2/610	Profile 40x40, 4E, light, plain	1.21.4E0, Connector, universal	2
10	1.11.040040.43LP-AA2AA2/610	Profile 40x40, 4E, light, plain	1.21.4E0, Connector, universal	4
11	1.11.040040.43LP-AA4AA4/460	Profile 40x40, 4E, light, plain	1.21.4E0, Connector, universal	6
12	1.11.040040.43LP-AA4AA4/610	Profile 40x40, 4E, light, plain	1.21.4E0, Connector, universal	6
14	1.11.040040.43LP-AA4AA4/520	Profile 40x40, 4E, light, plain	1.21.4E0, Connector, universal	8
19	2.87.1571-99	Polycarbonat 4mm	0.63.D07991.04012, Flat head screw, DIN 7991 - M4x12 1.31.EM4, Threaded plate E, M4	10 10
20	2.87.1571-99	Polycarbonat 4mm	0.63.D07991.04012, Flat head screw, DIN 7991 - M4x12 1.31.EM4, Threaded plate E, M4 0.63.D00912.08016, Cylindric head screw, DIN 912 - M8x16	15 15 4
21	2.87.1571-99	Polycarbonat 4mm	0.63.D07991.04012, Flat head screw, DIN 7991 - M4x12 1.31.EM4, Threaded plate E, M4	24 24
23	1.45.31075	Swivel castor stop fix with bolt hole fitting, dia: 75	1.35.1141015, Threaded insert M14/M10, L15 0.63.D06912.10030, Cylindric head screw, DIN 6912 - M10x30	4 4
24	1.46.115	Angle 25x40, dia: 8.7	0.61.D06923.08, Hexagonal nut with flange DIN 6923 - M8 0.63.D06912.06014, Cylindric head screw, DIN 6912 - M6x14 0.63.D00912.08016, Cylindric head screw, DIN 912 - M8x16 1.32.4EM6, T-Nut for subsequent insertion, w. spring E. M6	4 4 4 4
25	1.61.210	Handle light	0.63.D06912.05035, Cylindric head screw, DIN 6912 - M5x35	4
26	1.65.2538078	Lock GD-ZN	0.63.D06912.06016, Cylindric head screw, DIN 6912 - M6x16 1.32.4EM6, T-Nut for subsequent insertion, w. spring E. M6	8 8
39	1.62.448.22/22	Hinge 48 fixed, A 22.5	1.32.4EM6, T-Nut for subsequent insertion, w. spring E, M6 0.63.D07991.06014, Flat head screw, DIN 7991 - M6x14	12 12

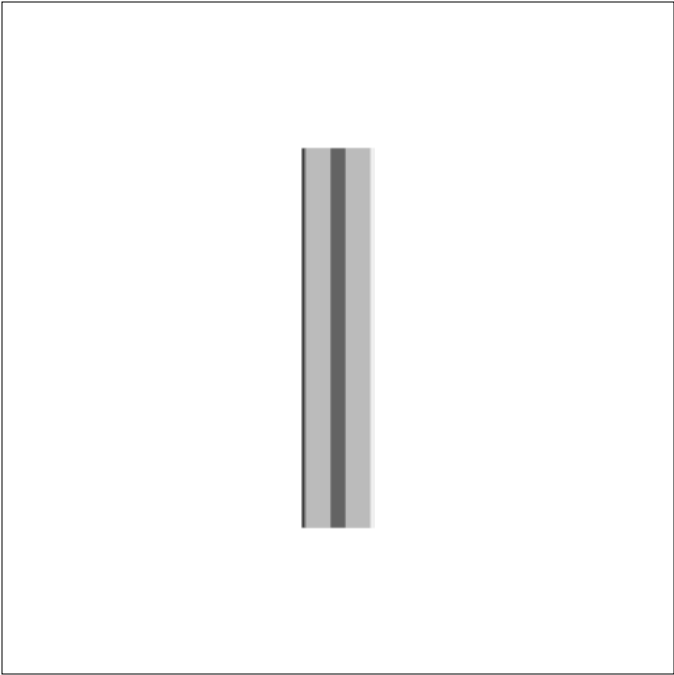
Profile machining details



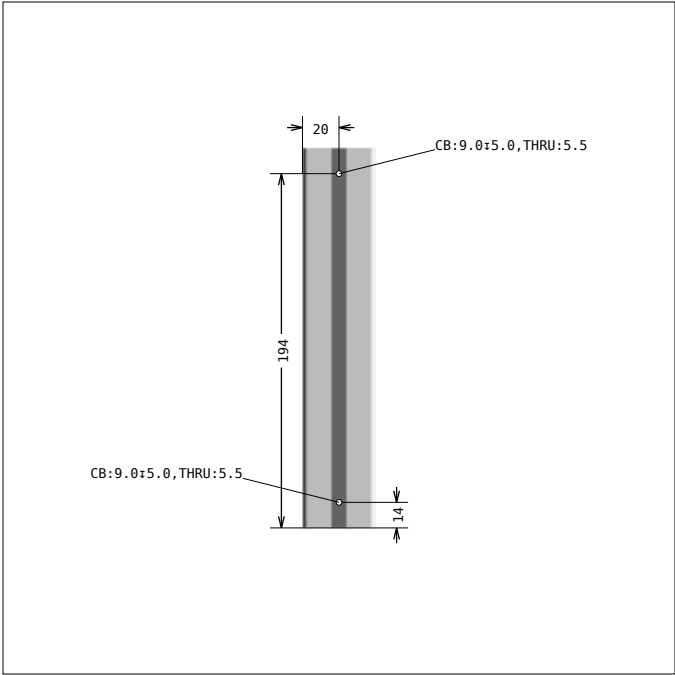
DRAWING 1, Pos 2
SIDE 1, bar length: 208.0 mm



DRAWING 1, Pos 2
SIDE 2



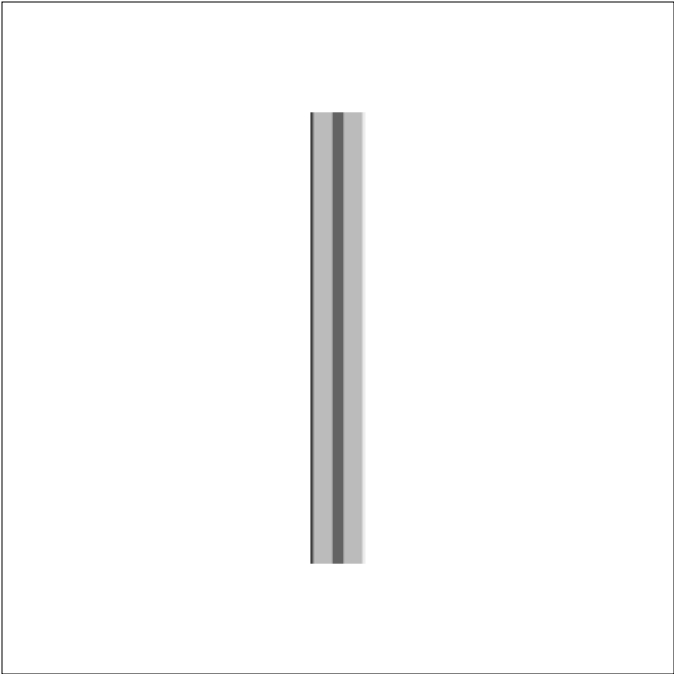
DRAWING 1, Pos 2
SIDE 3



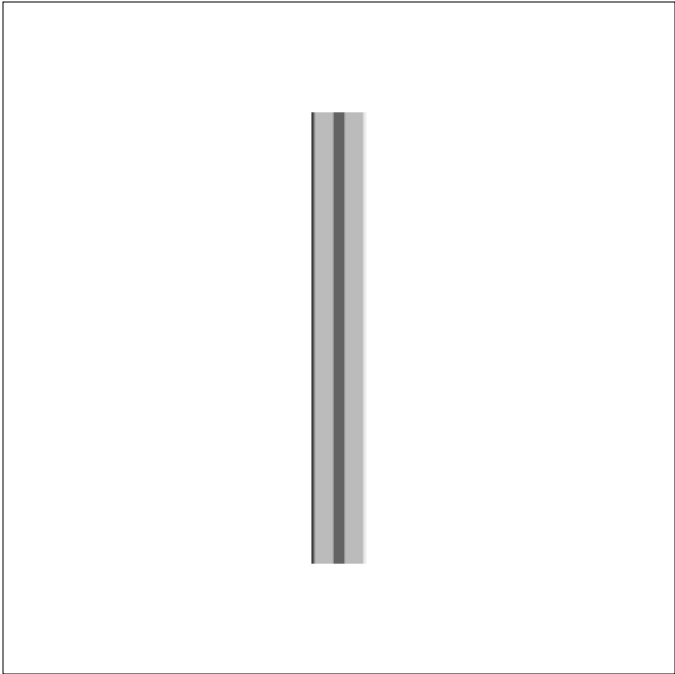
DRAWING 1, Pos 2
SIDE 4

← Profile "left end" is always at the bottom of the drawing. All dimensions are from "left".

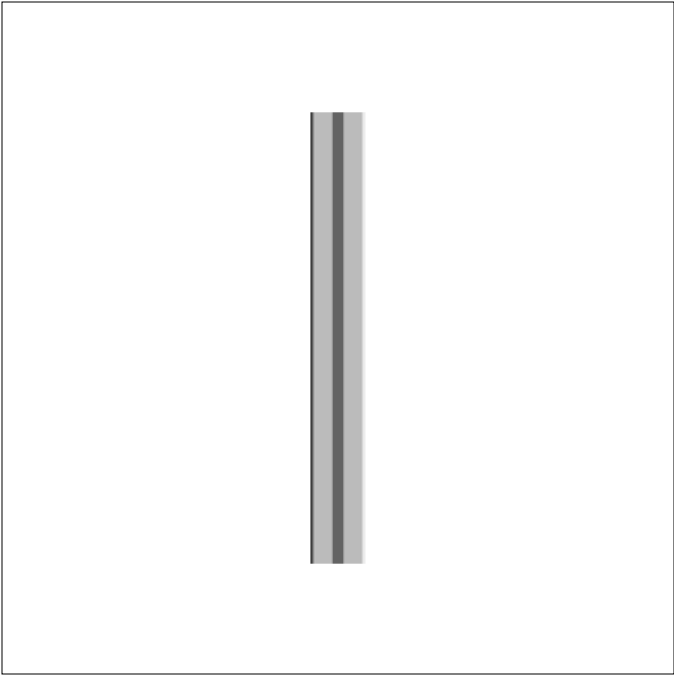
Profile machining details



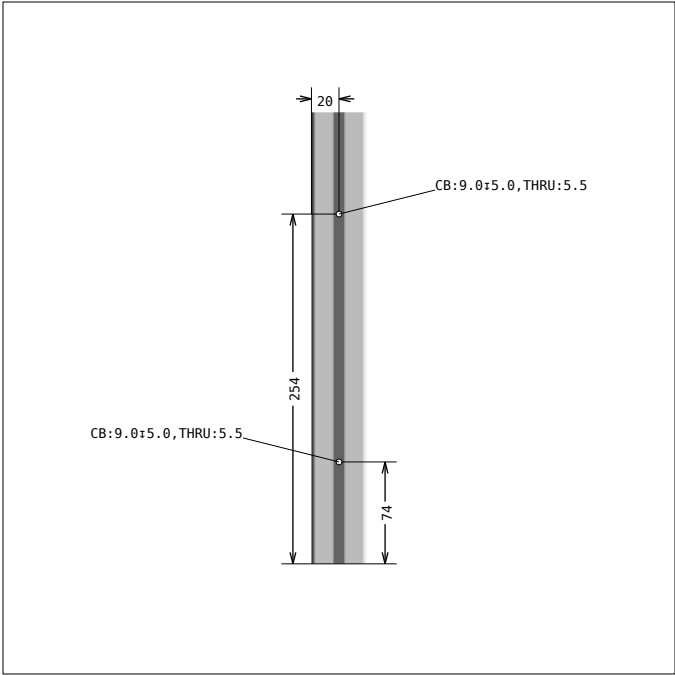
DRAWING 2, Pos 4
SIDE 1, bar length: 328.0 mm



DRAWING 2, Pos 4
SIDE 2



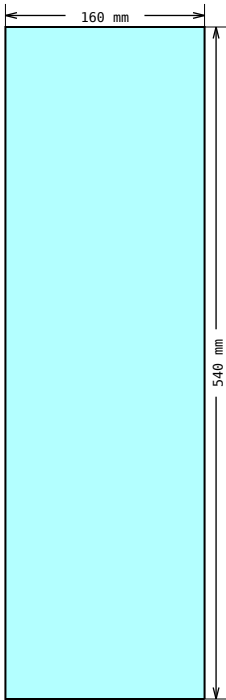
DRAWING 2, Pos 4
SIDE 3



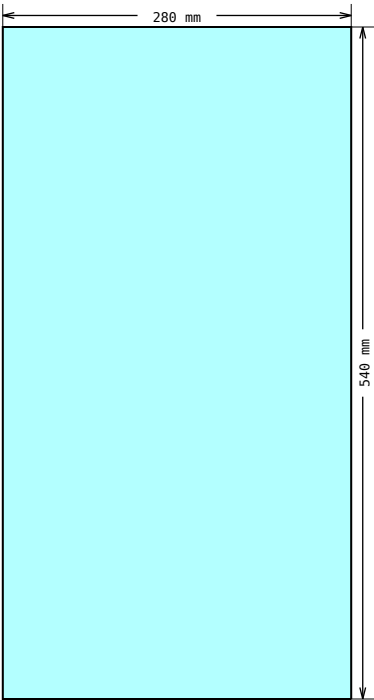
DRAWING 2, Pos 4
SIDE 4

← Profile "left end" is always at the bottom of the drawing. All dimensions are from "left".

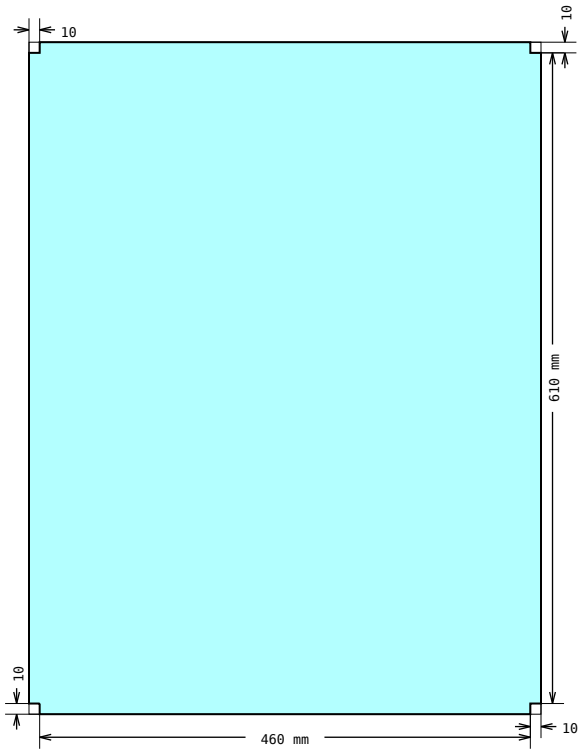
Panels



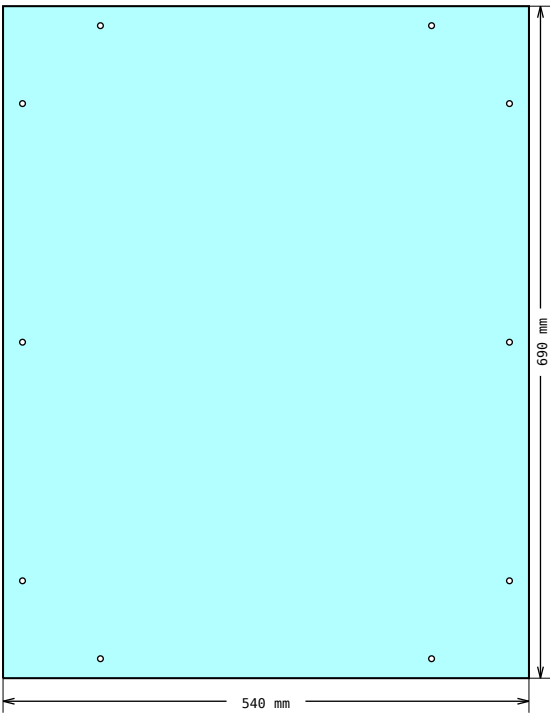
DRAWING 3, Pos 16
Shape measurements



DRAWING 4, Pos 17
Shape measurements

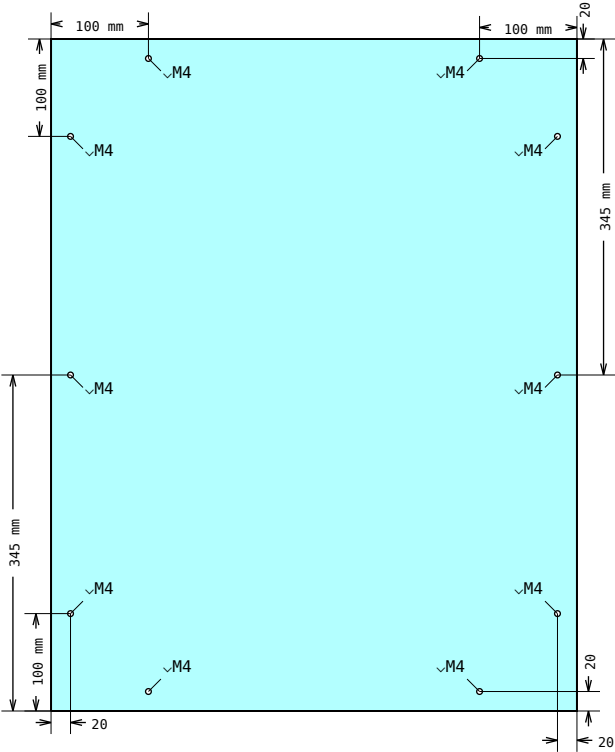


DRAWING 5, Pos 18
Shape measurements

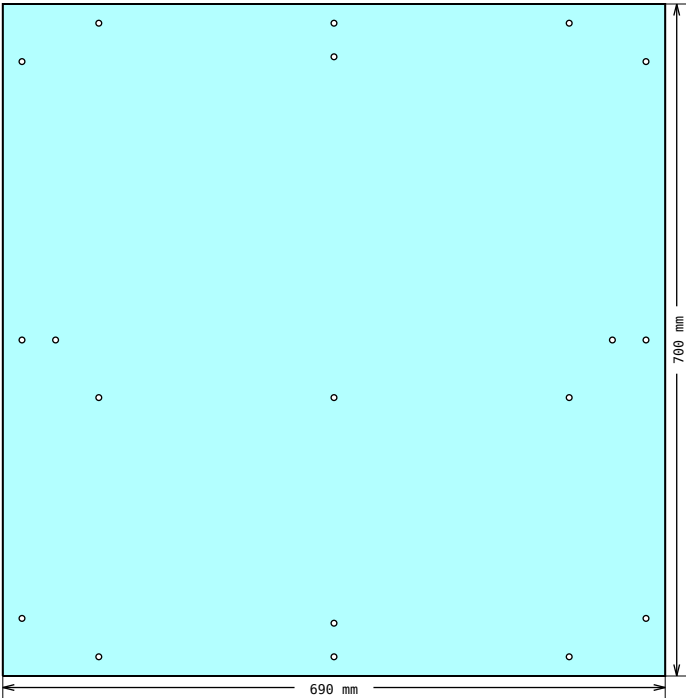


DRAWING 6, Pos 19
Shape measurements

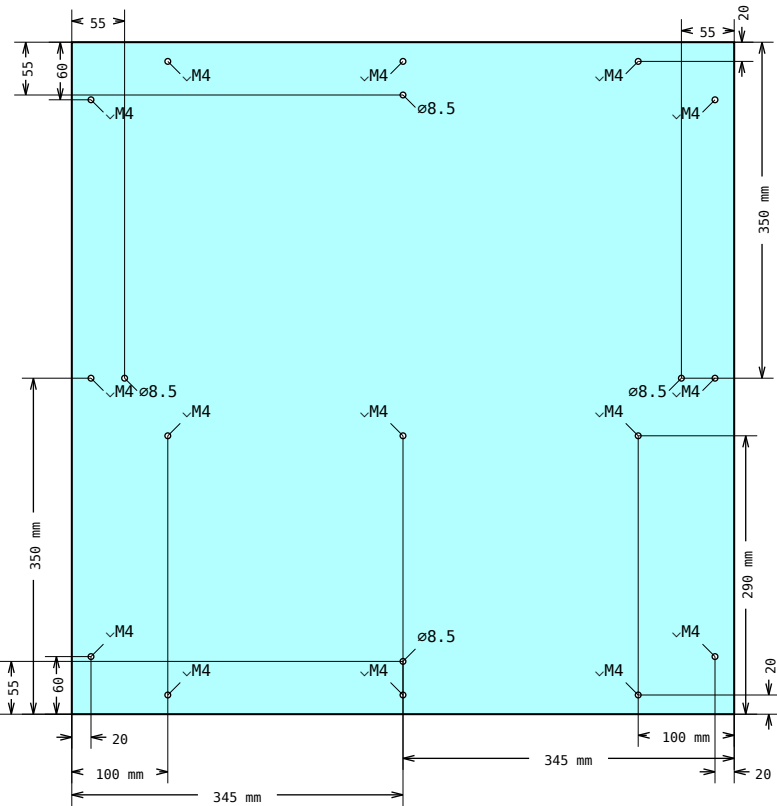
Panels



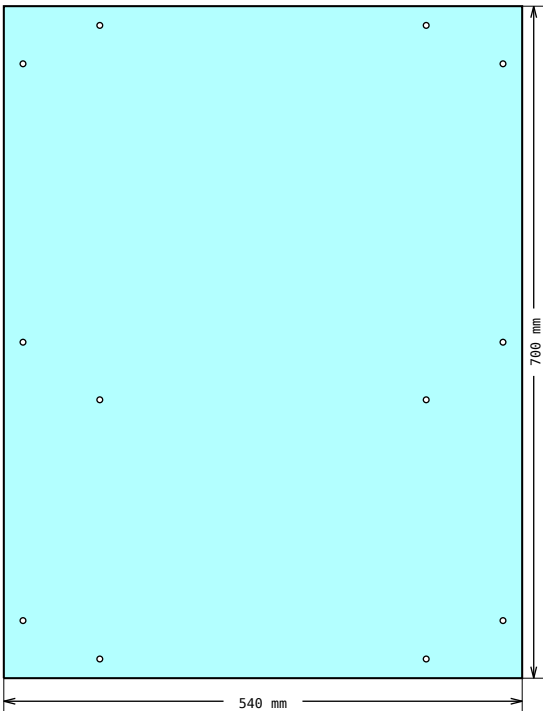
DRAWING 6, Pos 19
Hole placement



DRAWING 7, Pos 20
Shape measurements

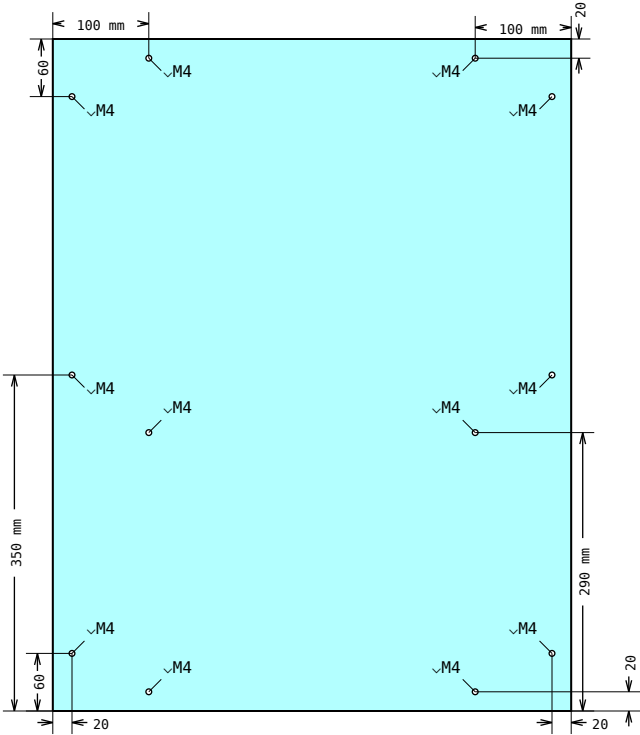


DRAWING 7, Pos 20
Hole placement



DRAWING 8, Pos 21
Shape measurements

Panels



DRAWING 8, Pos 21
Hole placement



Connectors for profiles with core hole-Ø 12 mm

1.21.4E0

Connection	Connector	Finished dimension	PG	Article-No. for connector with E-head		
				steel standard	E	VA
Universal			40	1.21.4E0	E	V

Connector types and materials

Technical data				
Connector	Article-No.	Material	Strength	Surface
Standard ground	1.21.4E0 1.21.4E0 E	steel	≥ 650 N/mm²	galvanised
Standard VA	1.21.4E0 V	stainless steel 1.4305	490 - 685 N/mm²	pickled and passivated

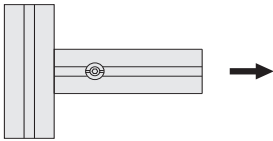
Strength values for profile connections with MayTec-Connectors

Torque tightening values for connector setscrew

PG	Slot	Setscrew special execution	Torque value	
			recommended	max.
40	E	M10×12	30.0 Nm	40.0 Nm

Comments
The max. tightening values are only valid for the MayTec setscrew and can not be reached by the usual commercial quality standard.

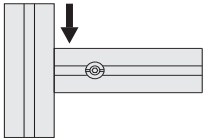
Tension load



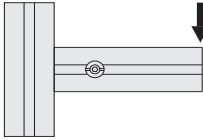
PG	Slot	Connector	max. Tensile strength
40	E	Universal	12,000 N

Comments
All values given have been tested with pretension of the connectors and maximum torque value and refer to the connection of two identical profiles.

Slide load



Flexure load

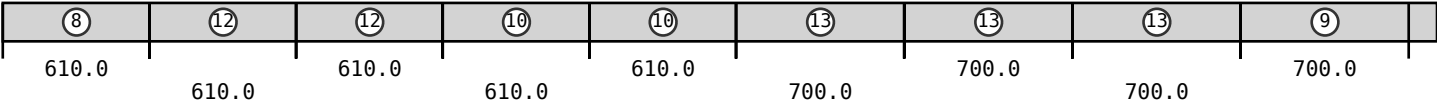
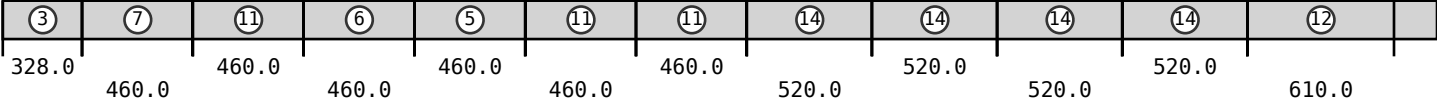


PG	Slot	Profile	Pcs	max. Slide strength		max. Flexure strength		
				Connector, Universal	E-connector	Connector, Universal		
40	E	40×40	1	6,000 N	9,000 N	250 Nm	500 Nm	1,000 Nm
		40×80	2	12,000 N	18,000 N			
		80×80, 8E	4	24,000 N	36,000 N			

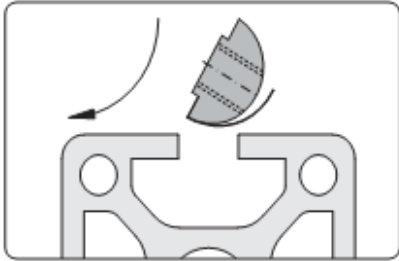
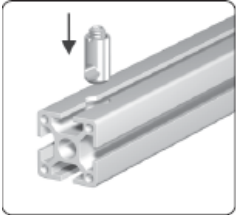
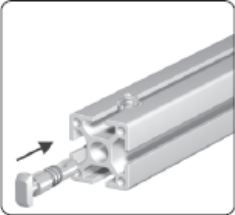
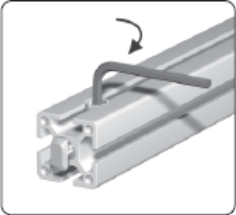
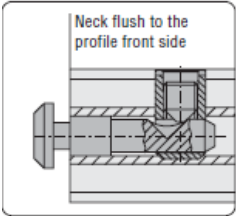
The listed values are valid for all light and heavy profiles

E = ground-connector, VA = stainless steel 1.4305

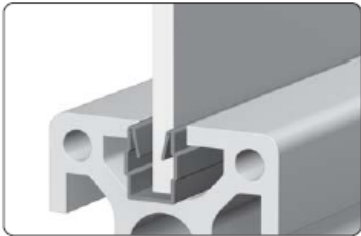
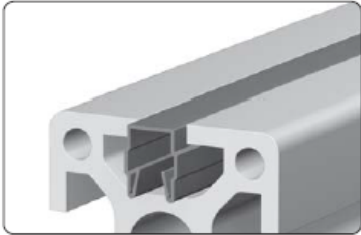
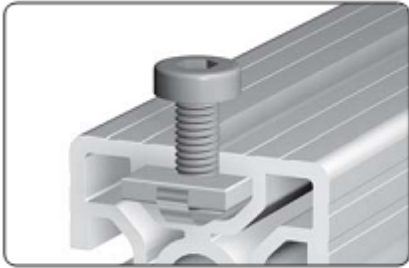
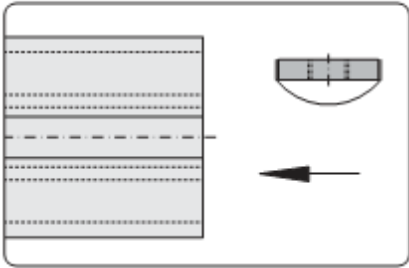
Suggested profile cuts (incl. kerf: 3.5 mm)

Article-No	Cuts	Waste (mm)
1.11.040040.43LP		118.5 mm
		180.0 mm
		5245.5 mm

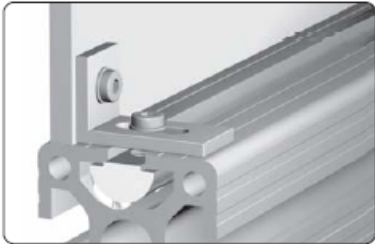
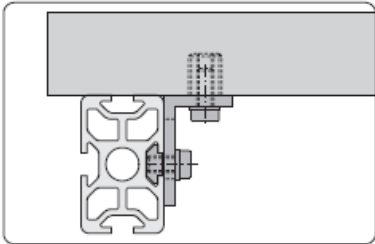
Assembly hints

ACCESSORY	1.32.4EM6	T-Nut for subsequent insertion, w. spring E, M6
 Fixing with leaf spring  Insert front-sided and rotate Application Fastening element for screw-type connections Technical data Design steel: • material: steel • surface: galvanised Design stainless: • material: stainless steel 1.4305 • surface: pickled and passivated max. moment of torque: $M_{A, max}$		These t-nuts are easily inserted into assembled frames
CONNECTOR	1.21.4E0	Connector, universal
 Insert the cross bushing  Push in the anchor  Pretension the anchor  Neck flush to the profile front side Comments For the optimal assembly of the profiles the connector is to be installed in such a way that the neck is flush to the profile front side		Most MayTec connectors follow these assembly steps. Tightening torque is: 5-15 Nm for 20mm profiles, 30 Nm for 30mm group, 35 Nm for 40mm and larger profiles.

Assembly hints

None	1.41.E314.1	Combination profile PP, E, 1 - 4.5, grey (bar 2.5 m)	
 Use as reduction profile  Use as slot-cover profile Application Combination profiles for use as reduction or cover profiles Technical data bar length: 2.5 m material: PP • oil and water resistant Colours: grey black orange red blue green			These profiles fill the space between the panel and the walls of the T-slot.
None	1.31.EM4	Threaded plate E, M4	
 Fixed into position with leaf spring  Assembly Insert from end Application Fastening element for screw-type connections			These plates must be inserted from the open side of the profile, prior to final frame assembly

Assembly hints

ACCESSORY	1.46.115	Angle 25x40, dia: 8.7
 Fastening of panels  Fastening of table tops		Application Angle bracket for the mounting of panels, table tops, switches and accessories These angles can be used to attach panels or thick tops.